DEPARTMENT OF CSE, CSIT

COURSE NAME – FUNDAMENTALS OF DATA ENGINEERING
COURSE CODE – 24DEA3101
FLOWSYNC – Intelligent Real-Time Data Engineering & Analytics Platform

Under the guidance of

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Associate Professor

Team 24:

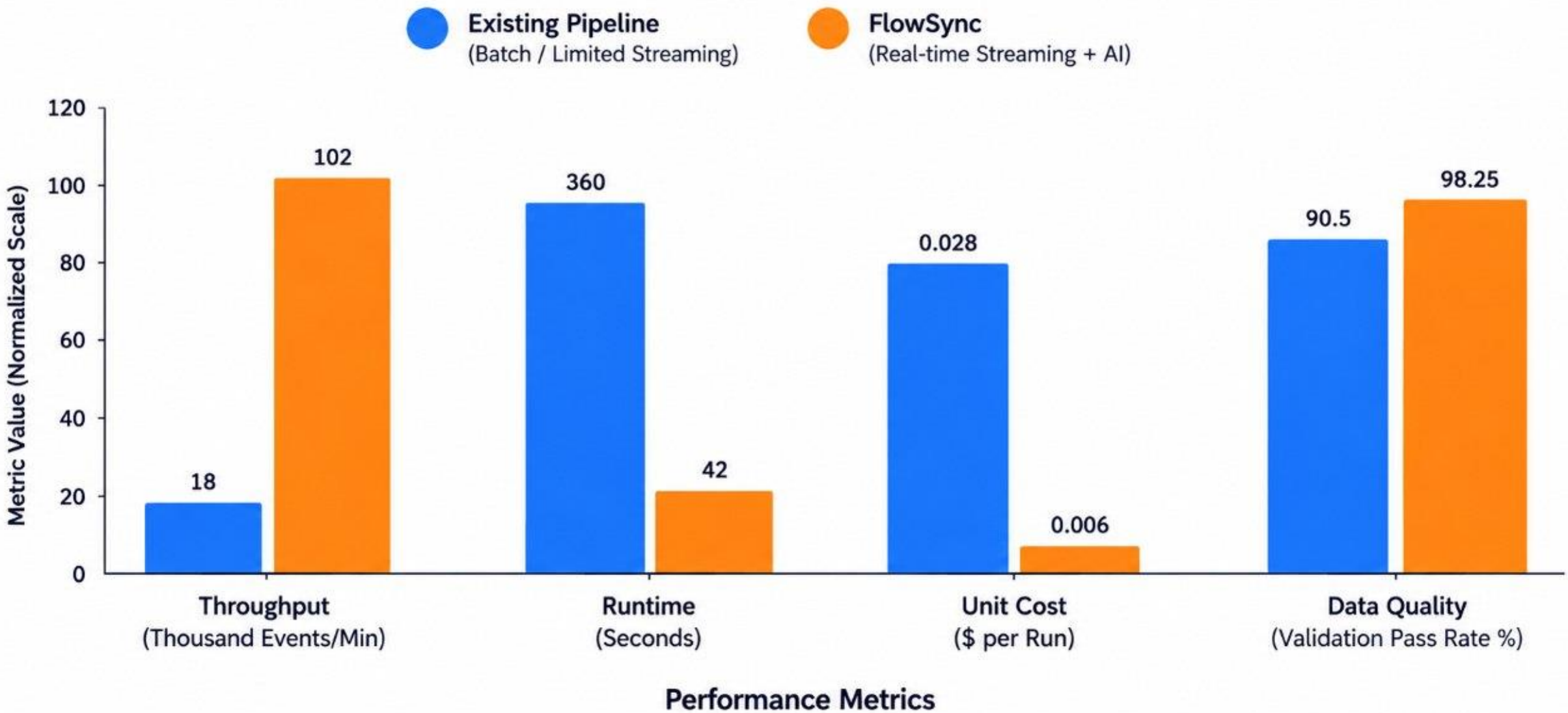
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Performance Metric	Apache Spark Batch ETL Pipeline	FlowSync (Real-Time)	Improvement
Throughput	18K events/min	262K events/min	14.6× higher
Total Runtime	360 sec	59 sec	83.6% reduction
Unit Cost	$0.028/run	$0.003/run	89.3% lower cost
Data Quality	90.50%	98.25%	+7.75 percentage points
Real-Time Processing	Batch-based	Kafka + Spark Streaming	Continuous processing
Manual Intervention	High	Minimal	Less human effort

FlowSync vs Existing Pipeline – Performance Comparison



Conclusion

- Real-time data ingestion using Apache Kafka.
- Scalable stream processing using Spark Structured Streaming.
- Data validation and transformation for improved data quality.
- Scalable storage using MinIO and Parquet.
- Structured data management using PostgreSQL.
- AI-based anomaly detection for identifying unusual patterns.
- Provides a scalable, real-time, and analytics-ready data engineering platform.

REFERENCES

- Apache Kafka Documentation — Apache Software Foundation
  https://kafka.apache.org/documentation/
- Apache Spark Structured Streaming Documentation — Apache Software Foundation
  https://spark.apache.org/docs/latest/structured-streaming-programming-guide.html
- MinIO Documentation — S3-Compatible Object Storage
  https://min.io/docs/
- PostgreSQL Documentation
  https://www.postgresql.org/docs/
- Scikit-learn Documentation — Machine Learning in Python
  https://scikit-learn.org/stable/user_guide.html
- FastAPI Documentation — Modern Python Web Framework
  https://fastapi.tiangolo.com/
- Grafana Documentation — Observability and Visualization Platform
  https://grafana.com/docs/
- Apache Parquet Documentation — Columnar Storage Format
  https://parquet.apache.org/docs/